

Feeling the Same, Seeing the Same:
Actual and Perceived Affect Similarity in Romantic Couples

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Abstract

Affect similarity in romantic couples is inconsistently associated with relationship outcomes, partly because studies rarely separate actual from perceived similarity, positive (PA) from negative affect (NA), or within-moment from between-dyad processes. In this preregistered dyadic study, we used a similarity index that combines dyadic elevation in baseline-centred affect with a mismatch penalty, capturing shared elevated affect rather than mere score proximity. 102 couples completed a 14-day experience sampling protocol (Study Component 1), and 98 couples returned for a structured laboratory protocol with video-mediated recall (Study Component 2). We tested associations between actual and perceived PA and NA similarity, momentary love, and post-interaction love and closeness, and examined the moderation of these associations by perceived partner responsiveness, event valence, and trait we-ness. Within moments, PA similarity predicted higher love and NA similarity lower love, with perceived similarity mediating roughly three-quarters of the PA association and two-fifths of the NA association. The mirror image dissolved at the dyad level, where couples whose negativity was chronically more aligned reported higher, not lower, love. Perceived partner responsiveness substituted for shared positivity and buffered shared negativity. Tests that PA similarity is most beneficial in negative moments and NA similarity most detrimental in positive moments were not supported with the preregistered event-valence index, which covered only partner-absent moments; extending the index to all moments instead pointed to amplification by negativity. PA and NA similarity are not mirror-image phenomena across all timescales, and perceived similarity operates more as a momentary mediator than as a stable interaction-level one.

Keywords: affect similarity, perceived similarity, romantic couples, experience sampling, dyadic data analysis

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Decades of work in social and personality psychology have posited that similarity between romantic partners benefits their relationship, fostering mutual understanding, closeness, and a sense of being kindred spirits (Anderson et al., 2003; Berlamont et al., 2024; Luo, 2017; Murray et al., 2002). Empirical findings, however, are mixed: across domains of similarity (e.g., traits, values, demographics, attitudes, and preferences), effects tend to be small and inconsistent (From et al., 2025; Luo, 2017; Tidwell et al., 2013). The picture is no clearer for affect similarity in romantic couples, understood as feeling the same as one's partner, for which associations with relationship evaluations such as satisfaction or closeness are equally inconclusive (Anderson et al., 2003; Gonzaga et al., 2007, 2010; Sels et al., 2020).

One possible explanation for these inconsistencies is that the effects of affect similarity may depend on the valence of the affect. Similarity in positive affect (PA) is generally believed to predict relationship satisfaction in dating and married couples (Gonzaga et al., 2007, 2010). However, other work has failed to find consistent benefits (From et al., 2025). Similarity in negative affect (NA), especially in conflict or stress contexts, has more often been associated with lower relationship quality (Brown et al., 2022; Gottman & Levenson, 2000). Conceptually, PA and NA similarity may matter for different reasons. Shared positivity may indicate mutual appraisals, interpersonal attunement, or effective co-regulation, all of which are plausible pathways to intimacy (Reis & Gable, 2015). In contrast, shared negativity may reflect co-rumination, escalation during interpersonal conflicts, or the transmission of stress between partners (Butler, 2015; Rose, 2002; Westman, 2001). Together, these arguments suggest that PA and NA similarity should be modelled separately and tested for opposing associations with relationship evaluations: a positive association for PA similarity and a negative one for NA similarity.

A second reason for inconsistent findings may be that theory and recent empirical work suggest it is not affective similarity itself, but the perception of such similarity that drives relationship outcomes (Montoya et al., 2008; Sels et al., 2020). What partners believe may matter as much as, or more than, what is objectively present. Because much of the existing research has measured either

actual or perceived similarity in isolation rather than jointly, the unique contribution of each remains unclear (From et al., 2025). Modelling both side by side allows a direct test of whether perceived similarity statistically accounts for the link between actual similarity and relationship evaluations. It motivates a formal mediation test of perceived similarity as the mechanism through which actual similarity relates to relationship evaluations.

Third, whether affect similarity is helpful or harmful likely depends on relational and contextual factors. Two candidates are particularly plausible. First, perceived partner responsiveness (the extent to which an individual feels understood, validated, and cared for by their partner) is a core relational resource that influences the extent to which interactions translate into intimacy (Reis & Gable, 2015). Shared positivity accompanied by high responsiveness may be experienced as joint engagement. In contrast, shared negativity in those same moments may be reframed as “being in this together” rather than as mutual distress. Second, situational valence may matter: shared positivity in a difficult context may be experienced as joint resilience, whereas shared negativity in an otherwise positive context may feel discordant. These two candidates motivate testing whether relational resources and situational valence moderate the similarity–relationship-evaluations link.

A possible final reason is that different studies quantify similarity in very different ways. Cronbach and Gleser (1953) argue in their classic measurement work that similarity indices can combine multiple components: elevation (mean level), mismatch (dispersion), and shape (profile pattern). Different operationalisations implicitly weight these components differently (Carlier et al., 2024). A pure distance score, for example, emphasises mismatch but is blind to whether partners are jointly high or jointly low, and a correlation captures shape but ignores elevation (Edwards, 2002; Nestler et al., 2019). This matters because many relationship theories are not about closeness in scores but about shared affective experiences of sufficient intensity (Reis & Gable, 2015). In the present study, we therefore conceptualise affect similarity as shared, co-experienced elevated affect: joint elevation in PA or NA relative to each partner’s baseline, coupled with low mismatch. The metric we use combines dyadic elevation with a penalty for absolute mismatch, resonating with conjunctive “weakest link” perspectives on group functioning (Steiner, 1972) and with mean-corrected-by-distance indices used in adjacent literatures, such as objective ambivalence (Schneider et al., 2015;

Thompson et al., 1995). This operationalisation thus reflects that what matters for relationship evaluations is shared affective experience of sufficient intensity, not closeness in scores.

Study Design and Hypotheses

Interpersonal theories of emotion dynamics emphasise that partners' emotions are not independent streams but can converge, react to one another, and become coupled through adaptive or maladaptive regulatory processes (Butler, 2015; Butler & Randall, 2013). This view implies that similarity is a fluctuating, time-varying process best captured by repeatedly measuring both partners in daily life or during structured interactions (Bolger & Laurenceau, 2013; Kenny et al., 2020). Study Component 1 of the present research, therefore, uses pre-existing experience sampling (ESM) data from a sample of romantic couples. A 14-day ESM protocol captured momentary affect, perceived partner affect, love, and contextual valence variables in daily life. In a complementary second study component, the same couples engaged in structured positive and negative laboratory interactions, after which their own and partner perceived affect during these interactions was reconstructed using video-mediated recall (VMR). Post-interaction questionnaires also probed for participants' feelings of love and closeness. Both partners additionally completed the Dyadic Coping Inventory (DCI; Bodenmann et al., 2018). Component 2 tests whether the daily-life patterns generalise to structured interaction contexts where the situational valence is fixed by design.

Table 1 shows the preregistered hypotheses (see <https://doi.org/10.17605/OSF.IO/E5SZ2> for the full preregistration). For Study Component 1, we first expect main effects of actual similarity on momentary love: higher actual PA similarity should be associated with higher love, both within and between dyads (H1.1), and higher actual NA similarity should be associated with lower love (H1.2).

When perceived similarity is added alongside actual similarity in the same model, we expect perceived similarity to predict love in the same direction as actual similarity and the actual-similarity coefficients to attenuate (H1.3). This attenuation pattern motivates our formal mediation tests, in which we expect the associations between actual similarity and love to be at least partially mediated by perceived similarity (H2.1 and H2.2).

We then examine two sets of moderators of the similarity-love link. We expect that higher momentary perceived partner responsiveness will amplify the positive PA association (H3.1) and

attenuate the negative NA association (H3.2). For event valence, we expect a mismatch pattern in which PA similarity is most beneficial in relatively negative moments (H3.3) and NA similarity is most detrimental in relatively positive moments (H3.4).

Table 1

Summary of Preregistered Hypotheses

	Study Component 1 (Daily Life) <i>Momentary measures of similarity and love, and of partner responsiveness and event valence as moderators.</i>	Study Component 2 (Lab Interactions) <i>Interaction-averaged measures of similarity, and of we-ness and interaction valence as moderators. Post-interaction love and closeness as outcomes.</i>
H1.1	Greater actual PA similarity is associated with higher love.	Greater actual PA similarity is associated with higher love/closeness.
H1.2	Greater actual NA similarity is associated with lower love.	Greater actual NA similarity is associated with lower love/closeness.
H1.3	Perceived similarity predicts love in the same direction as actual similarity, over and above actual similarity, and the actual similarity coefficients should attenuate.	Perceived similarity predicts love in the same direction as actual similarity, over and above actual similarity, and the actual similarity coefficients should attenuate.
H2.1	Perceived PA similarity mediates the H1.1 link.	Perceived PA similarity mediates the H1.1 link.
H2.2	Perceived NA similarity mediates the H1.2 link.	Perceived NA similarity mediates the H1.2 link.
H3.1	Perceived responsiveness amplifies the positive association between PA similarity (actual and perceived) and love.	We-ness amplifies the positive association between PA similarity (actual and perceived) and love and closeness.
H3.2	Perceived responsiveness attenuates the negative association between NA similarity (actual and perceived) and love.	We-ness attenuates the negative association between NA similarity (actual and perceived) and love and closeness.
H3.3	Actual/perceived PA similarity will be most beneficial for love in negative moments.	Actual/perceived PA similarity will be most beneficial for love/closeness in the negative laboratory interaction.
H3.4	Actual/perceived NA similarity will be most detrimental for love in positive moments.	Actual/perceived NA similarity will be most detrimental for love/closeness in the positive laboratory interaction.

Note. In Study Component 1, all hypotheses distinguish within-level (momentary) from between-level (stable) associations. Actual similarity is decomposed at the dyad level (within- and between-dyad)

and perceived similarity at the person level (within- and between-person), as detailed in *Methods*. PA = positive affect; NA = negative affect.

For Study Component 2, we expect parallel main effects of interaction-averaged similarity on post-interaction love and closeness (H1), with the same mediation logic via perceived similarity (H2). We additionally test two moderators: whether trait we-ness, a person-level index of shared dyadic coping derived from the DCI (see *Measures*), serves as a relational resource that amplifies the association between similarity and post-interaction love and closeness (H3.1, H3.2), and whether the lab interaction context (positive vs. negative discussion) produces the same mismatch pattern as event valence in daily life (H3.3, H3.4).

Method

Participants and Procedure

The data were collected as part of the larger Re-Connect project (for more information, see <https://ppw.kuleuven.be/reconnect>). 104 couples (208 individuals) were recruited through psychiatry and psychotherapy clinics, as well as through flyers distributed on social media and in public and clinical settings in Belgium and the Netherlands. Inclusion criteria were being at least 18 years old, being in a relationship for at least two months, living together, not residing in a treatment facility, being fluent in Dutch, and both partners owning a smartphone. The full project included a structured clinical screening, background questionnaires, a laboratory session with positive and negative interactions and VMR, and a 14-day ESM protocol. Compensation depended on compliance and ranged up to €100 per person. The study protocol was approved by the ethics committee, and all participants gave written informed consent.

For Study Component 1, we excluded one couple due to technical issues that made it unclear which responses belonged to which partner, and another couple in which one partner had an ESM compliance ratio below .30 (the preregistered cut-off). The final ESM sample thus comprised 102 couples (204 individuals). Participants ranged in age from 19 to 63 years ($M = 32.0$, $SD = 9.70$). Of the 204 individuals, 112 (54.4%) identified as female and 92 (44.7%) as male. The sample was predominantly Belgian (94.2%), with smaller proportions from the Netherlands (3.4%) and other countries (2.4%). Most participants identified as White (97.1%). Education levels were relatively

high: 36.9% held a master's degree, 32.5% a bachelor's degree, 26.7% had completed secondary education, and the remainder (3.9%) held either a lower or a doctoral degree. Couples had been in their relationship for 8 to 417 months ($M = 88.7$ months, $SD = 81.23$; median = 62 months), and nearly all were cohabiting (99%). A quarter of participants (24.8%) had children. Given the clinical recruitment channel, a substantial proportion of participants (79.6%) reported current medication use; this characteristic is reported for transparency but was not an exclusion criterion and was not entered as a covariate in the planned analyses.

For Study Component 2, we excluded couples who did not complete the lab session, leaving 98 couples (196 individuals), with similar characteristics as the Study Component 1 participants.

During the ESM period, partners received notifications five times per day on weekdays (between 5:00 p.m. and 10:15 p.m.) and ten times per weekend day (between 10:45 a.m. and 10:15 p.m.), to maximise the chance that they would be together. At each beep, both partners answered a brief questionnaire (approximately 2 minutes) about their current feelings, behaviours toward their partner, and their perceptions of their partner's feelings. Partners received beeps simultaneously.

In the laboratory session, couples first discussed events of the day for 2 minutes, then engaged in two 8-minute interactions: a positive interaction in which they discussed a topic both partners experienced positively, and a negative interaction in which they discussed a topic of disagreement. Topics were selected from the Problem Areas Questionnaire (Heavey et al., 1995) based on both partners' ratings. After each interaction, both partners independently completed a brief post-interaction questionnaire in private cubicles. Three cameras recorded each interaction. Both partners then watched the recording in a VMR procedure: every 30 seconds the recording was paused and partners reported their own and their partner's feelings during the preceding interval.

Measures

Momentary love (Study Component 1). At each beep, participants answered "How loving do you feel towards your partner at the moment?" on a slider from 0 (*not at all loving*) to 100 (*very loving*). Love is the relationship evaluation outcome for Study Component 1.

Post-interaction love and closeness (Study Component 2). After each interaction, participants rated love using the same item and a 0-100 slider scale. Closeness was assessed with the

item "How close do you feel to your partner right now?" on the same scale. Both outcomes are relationship-evaluation dependent variables for Study Component 2.

Affect. In the ESM, participants reported how positive and how negative they felt at the moment, and how positive and how negative they thought their partner felt at the moment, all on 0–100 sliders. In the VMR, participants used the same 0–100 sliders to report their own affect, their partner's affect, and their perception of their partner's affect during each 30-second segment.

Perceived partner responsiveness. In the ESM, participants reported the extent to which they felt understood by their partner at the moment on a 0–100 slider.

Event valence. In the ESM, participants indicated whether something positive and whether something negative had happened since the previous beep (1 = no, 2 = yes, a little, 3 = yes, a lot). These general event items were, by design, presented only in the branch questionnaire of the survey in which participants reported being neither currently with their partner nor in recent contact with them. For the preregistered tests of H3.3 and H3.4, we recoded these items on a 0–2 scale and combined them into a momentary context index, $C = \text{positive event} - \text{negative event}$ (with a possible range of -2 to 2), with higher values indicating relatively more positive contexts. This index thus carries an important measurement caveat: it estimates the event valence $\times$ similarity interaction exclusively on partner-absent, contact-free moments, which is a specific and relatively narrow slice of daily life, and arguably one in which momentary similarity–love dynamics are least likely to be salient, because partners are separated and events cannot be immediately processed together. To address this selection issue, we additionally constructed an exploratory extended index C^{expl} that, in the complementary branches of the beep tree, drew on partner-specific event items (asking whether something positive or negative had happened between the partners), thereby preserving the conceptual definition of momentary valence as the net balance between positive and negative events while recovering a substantially larger portion of the beep-level data.

Trait we-ness. Trait we-ness was constructed from the Dutch version of the Dyadic Coping Inventory (Bodenmann et al., 2018; Ponnet, 2012) as the mean of 22 items reflecting constructive stress communication and supportive dyadic coping (items reflecting negative or hostile coping were

excluded; see Appendix D). Internal consistency of the we-ness composite was good (Cronbach's $\alpha = .83$, McDonald's $\omega = .90$). Trait we-ness was used in Study Component 2 only.

Affect Similarity Indices

We conceptualised affect similarity at a given moment as shared, co-experienced elevated affect. Operationally, we defined similarity as dyadic elevation (the mean of the two partners' baseline-centred affect ratings) minus a penalty for mismatch (the absolute difference between the partners' ratings). For actual similarity, $S^{act} = (X + Y^{act})/2 - |X - Y^{act}|$, where X and Y^{act} are the actor's own affect and the partner's self-reported affect at that moment. For perceived similarity, the partner term was replaced by the actor's perception of the partner's affect (Y^{perc}). Higher values reflect a joint, aligned elevation in affect, while lower values reflect either jointly low affect, a mismatch, or both. The two PA and NA similarity indices were computed separately. Conceptually, this metric distinguishes shared low from shared high (which a pure distance score does not) and explicitly penalises mismatch.

In Study Component 1, similarity indices were decomposed into within and between components, following standard practice for separating within from between effects in multilevel models (Bolger & Laurenceau, 2013; Enders & Tofighi, 2007). Because actual similarity is symmetric within a dyad, whereas perceived similarity is actor-specific, the decomposition is defined at the dyad level for actual similarity (within- and between-dyad) and at the person level for perceived similarity (within- and between-person). Within components capture momentary deviations from a dyad's or person's typical level, while between components capture stable differences. In Study Component 2, the segment-level similarity indices were averaged across segments within each interaction to yield interaction-level similarity scores.

Analyses

All analyses followed the preregistered plan and were estimated in R (version 4.5.2) using the nlme package (Pinheiro et al., 1999), which allows for nested random effects and flexible residual covariance structures. Because our research questions do not distinguish between partners' roles, we treated partners as indistinguishable (Kenny et al., 2020).

For Study Component 1, all models were multilevel with moments nested within persons nested within dyads (Bolger & Laurenceau, 2013; Kenny et al., 2020), with random intercepts at the dyad and person levels and a continuous-time first-order autoregressive [CAR(1)] residual structure within persons to account for temporal dependence in the ESM data (Hamaker et al., 2015). The CAR(1) structure uses actual timestamps (in hours). The correlation between any two residuals separated by Δt hours is modelled as $\phi^{\Delta t}$ where ϕ estimates the correlation at a one-hour lag and governs how quickly autocorrelation decays with increasing elapsed time between beeps. Beep intervals varied due to missed beeps, nighttime gaps, and protocol variation, making continuous-time AR(1) more appropriate than discrete-lag AR(1). The primary models did not include random slopes in order to prioritise parsimony and reduce the risk of convergence problems in more complex models (Hox et al., 2017). Random slopes and other sensitivity analyses (time trend, elevation control, love-ceiling restrictions) were estimated separately and are reported in Appendix A.

Study Component 2 models were simpler in structure: persons were nested within dyads, with segment-level similarity indices aggregated over each interaction. Because each model was estimated separately within one interaction context, this left a single observation per person and models for H1, H2, and the trait we-ness moderation (H3.1 and H3.2) thus included a random intercept at the dyad level only. The context moderation models (H3.3 and H3.4) pooled both interaction contexts, yielding two observations per person, and therefore reintroduced a random intercept at the person level (nested within dyads).

Across both components, indirect effects in the mediation analyses (H2) were tested using Monte Carlo confidence intervals based on the estimated paths and their standard errors (Bauer et al., 2006; Preacher et al., 2010; Sels et al., 2020), which avoids the normality assumption for the product of coefficients and is less computationally demanding than bootstrapping in complex multilevel models. Furthermore, continuous time-varying predictors were person- or dyad-mean centred as appropriate, so that momentary effects reflect deviations from a person's or couple's usual level. Corresponding person- or dyad-level averages were grand-mean centred in order to estimate the between-person or between-dyad associations (Aiken & West, 1991; Enders & Tofighi, 2007).

Throughout, we used $\alpha = .05$ for inference, reported 95% confidence intervals, and did not correct for multiple comparisons, since each model tests a distinct hypothesis (García-Pérez, 2023).

As an exploratory complement, we conducted response surface analyses (RSA; Edwards, 2002; Nestler et al., 2019) to examine whether two structural assumptions underlying the preregistered similarity index hold when each partner's affect is entered as a free predictor in a second-degree polynomial model: that the decomposition into elevation and distance components adequately captures the functional form of similarity effects and that mismatch is symmetrically penalising. Because RSA was not preregistered and addresses a secondary question about the functional form of similarity effects, full details and results are reported in Appendix C.

Transparency and Openness

The study protocol, hypotheses, and analysis plan were preregistered on the Open Science Framework prior to data analysis (<https://doi.org/10.17605/OSF.IO/E5SZ2>). The repository also contains all analysis code, including both the preregistered models and the non-preregistered exploratory analyses reported here and in the Appendices, as well as a fully reproducible R environment with a step-by-step replication guide. An overview of the project materials, including rendered analysis outputs, presentations, summaries, the preregistration, and the project repository, is available through https://helianthos.github.io/Affect_Similarity/, the project's GitHub Pages portal.

Results

Study Component 1: Affect Similarity in Daily Life

Main Effects of Affect Similarity (H1.1–H1.3)

Table 2 presents the analytic sequence for Study Component 1. Models 1a and 1b tested the main effects of momentary actual and perceived affect similarity on momentary love (H1.1–H1.3). Model 1c is a preregistered exploration that adds the actual $\times$ perceived interaction term. A summary of results for these models can be found in Table 3.

Main effects of actual similarity (H1.1, H1.2). In Model 1a, consistent with H1.1, momentary actual PA similarity predicted momentary love: at moments when partners were more PA-similar than usual for that couple, both partners reported higher love ($\beta_1 = 0.11$, 95% CI [0.10, 0.12], $p < .001$). Contrary to H1.1, the between-dyad coefficient was small and not significant ($\beta_2 = 0.21$, 95%

CI $[-0.31, 0.74]$, $p = .42$). Consistent with H1.2, momentary actual NA similarity predicted lower love: at moments when partners were more NA-similar than usual, they reported lower love ($\beta_1 = -0.15$, 95% CI $[-0.17, -0.14]$, $p < .001$). Contrary to H1.2, however, the between-dyad coefficient was significantly *positive*: dyads who were on average more NA-similar across the ESM period reported *higher*, not lower, love ($\beta_2 = 0.62$, 95% CI $[0.20, 1.05]$, $p = .004$). We return to this between-dyad reversal in the Discussion.

Table 2

Overview of the Analytic Sequence for Study Component 1

Analysis	Model	Description	Affect	Hypothesis
Main effects of affect similarity	Model 1a	Actual affect similarity $\rightarrow$ love	PA	H1.1
			NA	H1.2
	Model 1b	Adds perceived similarity to Model 1a (no interaction)	PA, NA	H1.3
	Model 1c	Adds actual $\times$ perceived interaction term to Model 1b	PA, NA	Preregistered exploratory
Mediation analyses	Model 2	Multilevel mediation: actual affect similarity $\rightarrow$ perceived affect similarity $\rightarrow$ love	PA	H2.1
			NA	H2.2
Moderation analyses	Model 3a	Moderation by perceived partner responsiveness	PA	H3.1
			NA	H3.2
	Model 3b	Moderation by event valence	PA	H3.3
			NA	H3.4

Note. PA = Positive Affect, NA = Negative Affect.

Adding perceived similarity (H1.3). Model 1b added perceived affect similarity to the Model 1a predictors. Because perceived similarity is actor-specific, it decomposes at the person rather than dyad level. Perceived PA similarity emerged as a substantially larger predictor of love at both the within-person and between-person levels (within-person: $\beta_3 = 0.15$, 95% CI $[0.14, 0.16]$, $p < .001$; between-person: $\beta_4 = 0.86$, 95% CI $[0.37, 1.35]$, $p < .001$). The within-dyad actual PA similarity coefficient attenuated from 0.11 to 0.03 (a 73% reduction) but remained significant ($p < .001$), while the between-dyad coefficient remained non-significant. The same pattern emerged for NA similarity: at the momentary level, perceived NA similarity (within-person $\beta_3 = -0.13$, $p < .001$) was a stronger negative predictor of love than actual NA similarity (within-dyad $\beta_1 = -0.09$, $p < .001$), and the within-dyad actual NA coefficient attenuated from -0.15 to -0.09 when perceived similarity was

included. At the dyad level, the unexpected positive coefficient for actual NA similarity disappeared; the effect instead emerged at the person level for perceived NA similarity ($\beta_4 = 0.89$, 95% CI [0.36, 1.43], $p = .002$): people who typically perceived more NA similarity with their partner also reported higher love.

Exploratory Model 1c further revealed that actual and perceived similarity interact in their associations with love at both levels. Within persons, the two forms mutually amplify each other: at moments when both actual and perceived similarity are elevated, love is higher than either predictor alone would suggest (PA: $\beta_5 = 0.001$, 95% CI [0.001, 0.002], $p < .001$; NA: $\beta_5 = -0.002$, 95% CI [-0.002, -0.001], $p < .001$). At the between level, the interactions were likewise significant (PA: $\beta_6 = 0.093$, 95% CI [0.024, 0.162], $p = .009$; NA: $\beta_6 = 0.091$, 95% CI [0.040, 0.143], $p = .001$), indicating that typical actual and perceived similarity were not independent in their association with love.

Table 3

Summary of Models 1a, 1b, and 1c Results

Predictor	Level	PA Similarity			NA Similarity		
		1a	1b	1c	1a	1b	1c
Actual	Within (β_1)	0.113***	0.032***	.045***	-0.152***	-0.094***	-0.087***
Similarity	Between (β_2)	0.214	-0.355	-0.310	0.621**	0.010	0.022
Perceived	Within (β_3)	-	0.149***	.173***	-	-0.127***	-0.122***
Similarity	Between (β_4)	-	0.859***	.209***	-	0.893**	1.061***
Act. × Perc.	Within (β_5)	-	-	.001***	-	-	-0.002***
Similarity	Between (β_6)	-	-	0.093**	-	-	0.091***

Note. Model 1a = actual similarity only (H1.1 for PA, H1.2 for NA); Model 1b = actual + perceived similarity (H1.3); Model 1c = actual + perceived + actual × perceived interaction. All models include random intercepts for persons nested in dyads and a continuous-time AR(1) residual structure. '-' = predictor not in model. β_1 to β_6 indicate the model coefficients referenced in the text. Bold = $p < .05$, ** $p < .01$, *** $p < .001$.

Mediation Analyses (H2.1–H2.2)

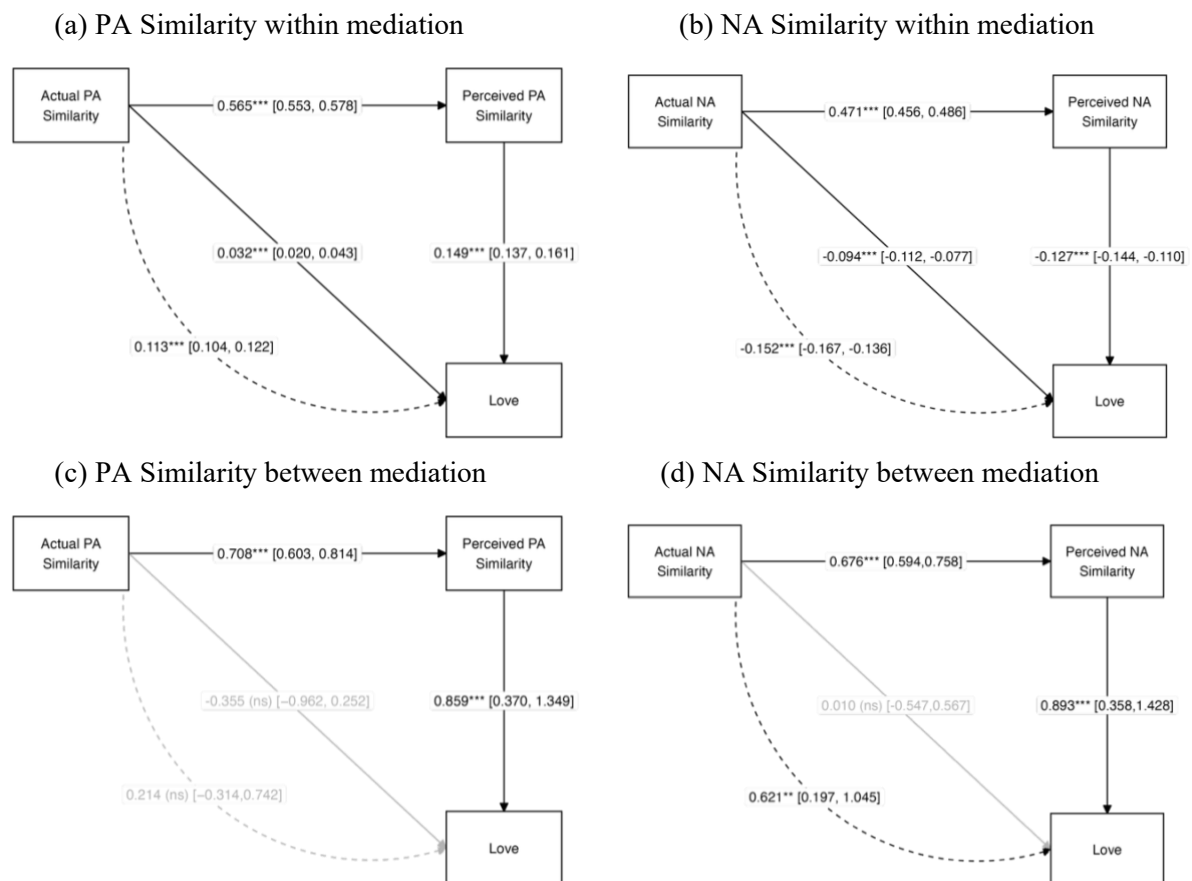
Model 2 tested the mediation of the actual similarity–love associations by perceived affect similarity (PA: H2.1; NA: H2.2). Results confirmed partial within-level mediation for both PA and NA, as shown in panels (a) and (b) of Figure 1. Indirect effects and 95% MC CIs are reported in Table

4. The within-level indirect effect for PA was proportionally larger than for NA (~75% vs. ~40% of the total within-level association mediated), suggesting a larger non-perceptual channel for NA.

At the between level, both PA and NA showed significant indirect effects with non-significant direct paths (panels c and d of Figure 1), indicating that perceived similarity statistically accounts for whatever between-level association exists with love. The two patterns differed only descriptively. For NA, full mediation emerged, while for PA, an inconsistent mediation pattern emerged: an albeit non-significant direct path ($\beta = -0.355$, $p = .249$) statistically cancelled the positive indirect effect, leading to a non-significant total effect (MacKinnon et al., 2000).

Figure 1

Mediation Diagrams for the Positive and Negative Affect Similarity–Love Associations



Note. Arrows indicate association and do not imply causation. Numbers on arrows are regression coefficients, followed by 95% CIs. Dashed arrows indicate total effects, taken from the H1.1/H1.2 models. Panels (a) and (b) chain momentary deviations; panels (c) and (d) chain typical levels (see Table 4 note for level structure). Indirect effects and proportion mediated are reported in Table 4. PA

= positive affect; NA = negative affect. *** $p < .001$. Greyed-out arrows and numbers indicate non-significant associations.

Table 4

Monte Carlo Indirect Effects for SCI Mediation Analyses (H2.1–H2.2)

	Level	Indirect effect	95% MC CI	p	Proportion mediated [95% MC CI]
PA Similarity	Within	0.084	[0.077, 0.092]	< .001	74.5% [65.8%, 81.0%]
	Between	0.609	[0.260, 0.972]	< .001	n/a (suppressed total effect)
NA Similarity	Within	−0.060	[−0.068, −0.052]	< .001	39.5% [33.5%, 44.8%]
	Between	0.604	[0.243, 0.974]	< .001	full mediation

Note. CIs based on 20,000 Monte Carlo draws. Within indirect effects link momentary deviations:

moments when a couple is more actually similar than usual, moments when a person perceives more similarity than usual, and momentary love. Between indirect effects refer to typical levels: couples' average actual similarity, persons' average perceived similarity, and persons' average love. Because actual similarity is at the couple level, whereas perceived similarity and love are at the person level, the indirect effects combine components across grouping levels and should be interpreted as statistical indirect paths rather than as clean single-level mediation effects. Proportion mediated not reported at the PA similarity between level where the total effect is non-significant due to suppression, as the ratio is undefined or unstable (MacKinnon et al., 2000). PA = positive affect; NA = negative affect.

Moderation Analyses (H3.1–H3.4)

Models 3a and 3b tested whether the within-person similarity–love associations were moderated by perceived partner responsiveness (H3.1, H3.2) and by the valence of recent events (H3.3, H3.4).

Moderation by perceived partner responsiveness (H3.1, H3.2). Table 5 summarises the Model 3a results for the four combinations of actual/perceived and PA/NA similarity. Perceived partner responsiveness was positively associated with momentary love (β s $\approx$ 0.34–0.37, $ps < .001$) and significantly moderated all four similarity–love associations, although not in the directions predicted across the board. For PA, contrary to H3.1, the positive association between similarity and love was *attenuated*, not amplified, at higher levels of perceived responsiveness, for both actual and perceived

PA similarity ($\beta_s = -0.002$, $ps < .001$). The PA similarity signal carried less weight when the partner was already perceived as responsive. For NA, consistent with H3.2, the negative association between similarity and love was *buffered* at higher levels of responsiveness, for both actual and perceived NA similarity ($\beta_s = 0.002$, $ps < .001$). Responsiveness thus appears to substitute for shared positivity but to buffer shared negativity, a functional asymmetry of the same relational resource.

Table 5

Perceived Partner Responsiveness as a Moderator of Similarity–Love Associations (H3.1–H3.2)

Affect	Similarity Type	Parameter	Estimate
PA Similarity (H3.1)	Actual	Similarity (within)	0.065***
		Similarity (between)	0.187
		Responsiveness	0.354***
		Similarity $\times$ Responsiveness	−0.002***
	Perceived	Similarity (within)	0.096***
		Similarity (between)	0.698**
		Responsiveness	0.339***
		Similarity $\times$ Responsiveness	−0.002***
NA Similarity (H3.2)	Actual	Similarity (within)	−0.072***
		Similarity (between)	0.618**
		Responsiveness	0.370***
		Similarity $\times$ Responsiveness	0.002***
	Perceived	Similarity (within)	−0.075***
		Similarity (between)	0.907***
		Responsiveness	0.368***
		Similarity $\times$ Responsiveness	0.002***

Note. Similarity (within): main effect of within-person/dyad similarity at average responsiveness.

Similarity (between): main effect of between-person/dyad similarity. Responsiveness: main effect of person-mean-centred perceived partner responsiveness. Similarity $\times$ Responsiveness: interaction term (preregistered hypothesis: positive for PA [H3.1] and NA [H3.2]). All models include random intercepts for persons nested in dyads and a CAR(1) residual structure. PA = positive affect; NA = negative affect. Bold = $p < .05$, ** $p < .01$, *** $p < .001$.

Moderation by event valence (H3.3, H3.4). In Model 3b, the preregistered tests using the momentary valence index *C* (see *Measures*) were all null: none of the four within-person similarity $\times$ event valence interactions reached significance (all $ps \geq .15$), although event valence itself had a substantial positive main effect on momentary love in every model ($\beta s \approx 2.2\text{--}2.8$, $ps < .001$).

Table 6

(Extended) Event Valence as a Moderator of Similarity–Love Associations (H3.3–H3.4)

Affect	Similarity	Parameter	Estimate
PA Similarity (H3.3)	Actual	Similarity (within)	0.077***
		Similarity (between)	0.200
		Event Valence (ext.)	4.779***
		Similarity $\times$ Event Valence (ext.)	−0.049***
	Perceived	Similarity (within)	0.122***
		Similarity (between)	0.702**
		Event Valence (ext.)	4.387***
		Similarity $\times$ Event Valence (ext.)	−0.037***
NA Similarity (H3.4)	Actual	Similarity (within)	−0.093***
		Similarity (between)	0.608**
		Event Valence (ext.)	4.991***
		Similarity $\times$ Event Valence (ext.)	0.055***
	Perceived	Similarity (within)	−0.098***
		Similarity (between)	0.899***
		Event Valence (ext.)	4.862***
		Similarity $\times$ Event Valence (ext.)	0.053***

Note. Exploratory extension of event valence moderation. Extended event valence (C^{expl}) = person-mean-centred net positivity pooled across general (partner-absent beeps) and partner-related (partner-present or partner-contact beeps) event items, covering a substantially larger portion of the data than the preregistered *C* index. Similarity (within): main effect at average event valence. Similarity (between): main effect of between-dyad similarity. Similarity $\times$ Event Valence (ext.): interaction with extended event valence. All models include random intercepts for persons nested in dyads and a CAR(1) residual structure. Bold = $p < .05$. ** $p < .01$, *** $p < .001$.

Re-estimating the four models with the exploratory extended index C^{expl} (see Table 6), all four similarity $\times$ event valence interactions reached significance (all $ps < .001$). The PA interactions were negative for both actual ($\beta = -0.049$) and perceived ($\beta = -0.037$) similarity, consistent with the preregistered mismatch expectation that PA similarity is most beneficial in less positive moments (H3.3). The NA interactions were positive and of comparable magnitude for actual ($\beta = 0.055$) and perceived ($\beta = 0.053$) similarity, running opposite to the H3.4 prediction and suggesting that NA similarity is less detrimental in more positive situations. Two further specifications using separate positive- and negative-event dummies, pooled across branches, yielded consistent results (see Appendix B). The preregistered null on event valence moderation should thus be read in light of the sampling restriction inherited from the ESM branching structure.

CAR(1) decay estimates. Across all primary Study Component 1 models, the estimated CAR(1) decay parameter ranged from $\phi = 0.31$ to $\phi = 0.48$ for outcome models (love as criterion) and from $\phi = 0.29$ to $\phi = 0.32$ for mediator models (perceived similarity as criterion), confirming that residual autocorrelation was present. Figure A1, panel (d) in Appendix A further shows that the CAR(1) structure was appropriate. Lower ϕ values in the mediator models are consistent with perceived similarity varying more freely across beeps than love.

Robustness and sensitivities. Diagnostic checks for the main-effects models (Models 1a and 1b) indicated non-normality and heteroscedasticity, both primarily reflecting the structural ceiling on the love measure (see Appendix A, where the full results are available and discussed). Because this concern stems from the bounded outcome and the shared multilevel structure, it applies equally to Models 1c, 2, 3a, and 3b. The preregistered sensitivity analyses (random-slope models, time-trend models, and models controlling for individual affect levels) and additional refits on ceiling-restricted subsamples yielded consistent fixed-effect estimates. These results are also discussed in Appendix A.

Study Component 2: Affect Similarity in Structured Lab Interactions

Study Component 2 followed the same analytic sequence as Study Component 1 (see Table 2), with two adaptations: (1) both post-interaction love and closeness served as outcomes, and (2) Models 3a and 3b tested trait we-ness and interaction valence rather than perceived partner responsiveness and event valence. Similarity was operationalised at the interaction level (averaged

across a structured 8-minute laboratory discussion) rather than the momentary level, and the situational valence was fixed by design (positive vs negative interaction topic) rather than assessed in the ESM stream.

Main Effects of Affect Similarity (H1.1–H1.3)

Models 1a and 1b tested the main effects of actual and perceived affect similarity on post-interaction love and closeness (H1.1–H1.3). Model 1c is an exploratory extension, adding the interaction between the two types of similarity. The model results can be found in Table 7.

Table 7

Summary of Study Component 2 Models 1a, 1b, and 1c Results

Outcome	Affect Similarity	Predictor	Positive Interaction			Negative Interaction		
			1a	1b	1c	1a	1b	1c
Love	PA Similarity	Actual	0.459**	0.386	0.393	0.379*	0.444*	0.437
		Perceived	-	0.105	0.099	-	-0.081	-0.067
		Actual × Perceived	-	-	0.003	-	-	0.003
	NA Similarity	Actual	0.352**	0.332	0.333	-0.141	-0.217	-0.200
		Perceived	-	0.025	0.026	-	0.108	0.096
		Actual × Perceived	-	-	0.001	-	-	-0.008
Closeness	PA Similarity	Actual	0.376*	0.215	0.221	0.432*	0.456*	0.464*
		Perceived	-	0.229	0.224	-	-0.031	-0.048
		Actual × Perceived	-	-	0.003	-	-	-0.004
	NA Similarity	Actual	0.383**	.528**	0.531**	-0.156	-0.175	-0.171
		Perceived	-	-0.177	-0.169	-	0.026	0.024
		Actual × Perceived	-	-	0.005	-	-	-0.002

Note. Model 1a = actual similarity only; 1b = actual + perceived similarity; 1c = actual + perceived + actual × perceived interaction. All models include a dyad-level random intercept. '-' = predictor not in model. Bold = $p < .05$. * $p < .05$, ** $p < .01$.

Main effects of actual similarity (H1.1, H1.2). In Model 1a, interaction-averaged actual PA similarity predicted higher post-interaction love in both interaction contexts: in the positive interaction, $\beta = 0.46$, 95% CI [0.13, 0.79], $p = .007$; in the negative interaction, $\beta = 0.38$, 95% CI [0.06, 0.70], $p = .020$. The PA similarity finding from daily life thus replicated cleanly in both

laboratory contexts, and the closeness models showed a parallel pattern (positive interaction: $\beta = 0.38$, $p = .016$; negative: $\beta = 0.43$, $p = .015$). For NA, the picture diverged from Study Component 1. In the negative interaction, actual NA similarity did not predict post-interaction love ($\beta = -0.14$, 95% CI $[-0.56, 0.28]$, $p = .504$). In the positive interaction, actual NA similarity was significantly and *positively* associated with post-interaction love ($\beta = 0.35$, 95% CI $[0.09, 0.62]$, $p = .009$), the opposite of what H1.2 predicted.

Adding perceived similarity (H1.3). In Model 1b, perceived similarity did not add unique predictive value beyond actual similarity in any model. High collinearity between interaction-averaged actual and perceived similarity ($r = .51-.73$) inflated standard errors, leaving neither predictor individually significant in most models. For closeness in the positive interaction, the actual NA coefficient strengthened from 0.38 to 0.53 when perceived NA was added, while perceived NA itself was non-significant ($\beta = -0.18$, $p = .20$).

Exploratory Model 1c added the actual $\times$ perceived interaction term to each of the 1b models. All interaction terms were statistically zero (all $ps > .54$), consistent across outcomes, affect types, and interaction contexts. This null pattern contrasts with Study Component 1, where significant interactions emerged at both levels for both PA and NA. Given that each Study Component 2 model rests on approximately 98 dyads, however, the absence of an interaction likely reflects limited statistical power rather than a genuine theoretical divergence.

Mediation Analyses (H2.1–H2.2)

As shown in **Error! Not a valid bookmark self-reference.**, Monte Carlo tests of the indirect effects of actual affect similarity on post-interaction love mediated by perceived similarity were uniformly non-significant across all Study Component 2 models. Decomposing the pathway clarified the source of this null. The association between actual and perceived similarity was strong in every interaction context with regression coefficients ranging from 0.70 to 0.82 (all $p < .001$), indicating that partners closely tracked each other's affective overlap. Once actual similarity was held constant, however, perceived similarity contributed no additional information about love or closeness in any context. The null indirect effects therefore reflect the absence of a unique evaluative contribution of perception, not a failure to perceive. H2.1 and H2.2 were not supported.

Table 8*Monte Carlo Indirect Effects for Study Component 2 Mediation Analyses (H2.1–H2.2)*

Affect Similarity	Context	Outcome	Indirect Effect	95% MC CI	<i>p</i>
PA (H2.1)	Negative	Love	−0.065	[−0.364, 0.235]	.677
		Closeness	−0.024	[−0.263, 0.284]	.858
	Positive	Love	0.074	[−0.161, 0.313]	.544
		Closeness	0.161	[−0.049, 0.435]	.151
NA (H2.2)	Negative	Love	0.076	[−0.209, 0.368]	.605
		Closeness	0.019	[−0.205, 0.336]	.891
	Positive	Love	0.020	[−0.207, 0.250]	.856
		Closeness	−0.145	[−0.311, 0.070]	.173

Note. CIs based on 20,000 Monte Carlo draws. Indirect effects represent the product of the a-path (actual similarity → perceived similarity) and b-path (perceived similarity → outcome), estimated within each interaction context. PA = positive affect; NA = negative affect.

Moderation Analyses (H3.1–H3.4)

Moderation by trait we-ness (H3.1, H3.2). Trait we-ness did not significantly moderate any of the PA similarity–outcome associations (H3.1; all interaction terms $ps \geq .23$). We-ness did predict love and closeness directly in the positive interaction context within the PA models (love: $\beta = 8.30$, 95% CI [1.41, 15.19], $p = .019$; closeness: $\beta = 10.93$, 95% CI [4.76, 17.10], $p < .001$), indicating that couples with a stronger sense of shared coping reported higher post-interaction evaluations after a positive exchange, independently of PA similarity.

For NA similarity (H3.2), no significant moderation emerged for actual NA similarity (all $ps \geq .56$). We-ness did predict love and closeness directly in the negative interaction context within the NA models (love: $\beta = 14.13$, 95% CI [5.87, 22.39], $p = .001$; closeness: $\beta = 15.21$, 95% CI [3.57, 26.84], $p = .011$), suggesting that couples with a stronger sense of mutual coping felt closer and more loving after a disagreement regardless of NA similarity. One significant moderation emerged: trait we-ness moderated the association between perceived NA similarity and closeness in the negative interaction ($\beta = 1.28$, 95% CI [0.41, 2.16], $p = .005$). Probing revealed a crossover: perceived NA similarity predicted less closeness at low we-ness ($\beta = -0.52$, $p = .021$) and more closeness at high

we-ness ($\beta = 0.54, p = .044$), with a statistically zero slope at average we-ness ($\beta = 0.009, p = .956$).

We-ness thus did not buffer against a negative perceived similarity effect as hypothesised but instead determined its direction: for couples with a strong sense of shared coping, perceiving mutual negativity during conflict may signal engagement rather than contagion.

Moderation by interaction valence (H3.3, H3.4). Context moderation was examined via pooled tests using a context dummy (0 = negative, 1 = positive). All context interactions were non-significant except one: actual NA similarity $\times$ context for love ($\beta = 0.55$, 95% CI [0.20, 0.89], $p = .002$). This means that H3.3 (on PA similarity interactions) is not supported. Probing the one significant interaction showed that actual NA similarity was significantly and negatively associated with love in the negative interaction ($\beta = -0.42$, 95% CI [-0.69, -0.15], $p = .002$), while the slope in the positive interaction was near zero and non-significant ($\beta = 0.12, p = .214$). H3.4 predicted that NA similarity would be more detrimental in the positive interaction (where shared negativity clashes with an otherwise pleasant context). However, the significant interaction ran in the opposite direction: NA similarity was harmful for love after the negative interaction but not in the positive one.

Discussion

This study examined how actual and perceived affect similarity in romantic couples is associated with relationship evaluations across two complementary contexts: a 14-day experience-sampling study of daily life (Study Component 1; SC1) and a structured laboratory protocol with video-mediated recall of momentary affect (Study Component 2; SC2). We organise the discussion around four findings. First, momentary positive affect (PA) and negative affect (NA) similarity show clear mirror-image associations with momentary love. However, this pattern dissolves and even reverses for NA at the dyad level, where couples with chronically more aligned negativity report higher, not lower, love on average. Second, perceived similarity captures most of the momentary similarity–love association, consistent with a perceptual mechanism, although this mediation does not survive aggregation across an 8-min interaction. Third, perceived partner responsiveness operates asymmetrically across the two valences, substituting for shared positivity and buffering shared negativity. Fourth, the preregistered mismatch account of contextual moderation fails on the partner-absent moments to which the original index was structurally restricted; once event valence extended

to the full beep set, contextual interactions emerge but mostly run opposite to the mismatch prediction.

A within-moment mirror image, dissolved between dyads

The mirror image of PA and NA at the within-moment level was the most robust pattern in the data. Moments at which a couple was more PA-similar than usual coincided with higher love for both partners; moments at which the couple was more NA-similar than usual coincided with lower love. This pattern is consistent with work on co-experienced affect in married couples, where shared positive affect was associated with better relationship quality and shared negative affect with poorer relationship quality (Brown et al., 2022). The finding also fits interpersonal dynamics perspectives in which partners' emotions become linked through convergence, reactivity, and interpersonal regulation (Butler, 2015), while echoing work on the negative side of emotional attunement, such as co-rumination and stress crossover (Rose, 2002; Westman, 2001). The within-moment pattern also extends ESM work in couples showing that momentary emotional similarity carries immediate relational meaning for perceived responsiveness and love (Sels et al., 2020).

A more interpretively demanding finding emerged at the dyad level for NA: couples whose ESM-period averages were more NA-aligned reported higher, not lower, momentary love on average. This between-dyad coefficient was fully accounted for by perceived NA similarity in the mediation model: dyads who tend to share negativity also tend to perceive each other as more similar, and it is the perceived alignment, not the affective alignment itself, that is associated with greater love. A directionally similar pattern surfaced in SC2: actual NA similarity during the positive laboratory interaction was positively, not negatively, associated with post-interaction love. Although the SC1 between-dyad coefficient and the SC2 interaction-averaged coefficient are not strictly comparable, both are aggregate rather than momentary, and both run counter to a within-moment account in which shared NA undermines love. We are cautious about reading this pattern causally, and the data do not arbitrate among plausible accounts. Chronically aligned negativity may indicate that partners feel understood by one another in times of distress, thereby building relational trust over time (Reis & Gable, 2015). Alternatively, shared negativity may be a byproduct of coping together: couples who face stress as a team share more negative experiences, but the relational bond comes from the joint

effort rather than from the negativity itself, in line with dyadic coping perspectives (Bodenmann et al., 2018). Third, the between-dyad association may reflect broader couple similarity rather than an effect of NA alignment itself: couples with more similar lives or trait dispositions may co-experience negative affect more often and feel like “kindred spirits”, with love reflecting this broader similarity rather than the negativity itself (Anderson et al., 2003; Luo, 2017; Murray et al., 2002). What the pattern does show is that NA similarity carries opposite signals at different timescales: aversive in the moment, affiliative when chronic.

The same scale asymmetry did not appear for PA. Within moments, PA similarity predicted higher love; between dyads, the actual-PA coefficient was small and uninformative after controlling for perceived similarity. This is consistent with the broader similarity literature, in which actual similarity shows mixed, often modest associations with relationship outcomes, whereas perceived similarity tends to be more consistently linked to relational well-being (From et al., 2025; Luo, 2017).

Perception is a momentary mechanism, but not an interaction-level one

H1.3 was supported in SC1: perceived similarity predicted love over and above actual similarity, and the within-dyad actual-similarity coefficients attenuated by roughly 70% (PA) and 40% (NA) when perceived similarity was added. The mediation analyses formalised this. About 75% of the momentary PA association and 40% of the momentary NA association ran through perceived similarity. The pattern adds daily-life evidence to the long-standing observation that what partners perceive about each other tends to predict evaluations more strongly than what is objectively the case (Montoya et al., 2008; Murray et al., 2002; Tidwell et al., 2013). The PA channel was found to be more perceptual than the NA channel: a larger residual within-moment NA association remained after accounting for perceived similarity, suggesting that shared negativity carries an evaluative signal that is not fully transmitted through conscious perception. One plausible interpretation is that part of the within-moment cost of shared NA is affective rather than inferential: simply being around a similarly distressed partner may pull the moment down due to stress or affect crossover, even when the perceived alignment accounts for only part of the association (Westman, 2001).

The mediation pattern did not generalise to the interaction-averaged scale. None of the SC2 indirect effects were significant. We see two non-exclusive reasons. First, averaging across an 8-min

interaction substantially reduces the within-dyad variance the mediator can carry. When both partners are summarised by a single number per interaction, the moment-to-moment perceptual updating that drives the SC1 indirect effect is no longer available. Second, the actual–perceived correlations in SC2 are high (.51–.73), so collinearity inflates standard errors and obscures partial effects. The SC2 null therefore does not undermine the SC1 mediation and is not necessarily evidence against perceptual transmission per se: in our data the evaluative role of perceived similarity appears to operate as a momentary process more than as a stable interaction-level one, as in earlier ESM work on actual and perceived emotional similarity in couples (Sels et al., 2020).

Responsiveness substitutes for shared positivity and buffers shared negativity

The pattern of responsiveness moderation in SC1 was internally consistent and asymmetric across valences, although not in the directions predicted across the board. Higher momentary perceived partner responsiveness attenuated the positive association between PA similarity (actual and perceived) and love, contrary to H3.1. It buffered the negative association between NA similarity and love, consistent with H3.2. Functionally, responsiveness behaved as a substitute for PA similarity and as a buffer for NA similarity.

The substitution pattern is consistent with relationship theory. If momentary shared positivity partly signals that the partner is engaged and attuned, then a partner who is already perceived as responsive provides that signal directly (Reis & Gable, 2015). There is less need for shared positivity to do additional work. Shared positivity is thus most informative for love when responsiveness is low; when responsiveness is high, the relational message is already received. The buffering pattern for NA similarity has a different logic. Shared negativity is plausibly more aversive when it occurs without felt understanding and is plausibly parallel suffering, which earlier work has linked to co-rumination dynamics and to stress crossover (Rose, 2002; Westman, 2001). When responsiveness is high, the same shared negativity may be reframed as being in this together, consistent with the view that responsiveness can foster, rather than undermine, intimacy and well-being (Reis & Gable, 2015).

Taken together, the pattern suggests that perceived responsiveness is a flexible resource whose function depends on affective context: redundant with shared positivity when partners already feel understood, and protective against shared negativity precisely when understanding is present.

This converges with broader perspectives on responsiveness as a flexible regulatory resource (Reis & Gable, 2015) and with empirical work showing that responsive partners enhance the relational benefits of positive disclosures and dampen the cost of negative ones (Gable et al., 2006; Maisel et al., 2008; Maisel & Gable, 2009).

Mismatch was not supported; amplification by negativity emerged in its place

The preregistered mismatch hypotheses (H3.3, H3.4) predicted that PA similarity would be most beneficial in negative moments and NA similarity most detrimental in positive moments. Using the original valence index, none of the four ESM interactions reached significance. The moderator's domain helps explain that null: the preregistered index could be calculated only at beeps in which participants reported neither being with their partner nor in recent contact with them, a structurally narrow slice of daily life and arguably the slice in which momentary similarity dynamics are least likely to be salient. When event valence was reconstructed from the full beep set in the exploratory alternatives, all four interactions emerged. For PA, the pattern was consistent with H3.3: PA similarity was most positively associated with love in less-positive moments. For NA, the pattern ran opposite to H3.4: NA similarity was more, not less, harmful when something negative had occurred.

SC2 was directionally consistent. In the pooled moderation, actual NA similarity predicted lower love after the disagreement and was uninformative after the positive interaction. Given SC2's much smaller effective sample size and the collinearity noted in the *Results*, we read this attenuated picture as consistent with the SC1 amplification pattern rather than contrary to it.

Across both components and event-valence operationalisations, NA similarity was most clearly harmful in negative contexts, extending prior work showing that co-experienced negative affect is associated with poorer relationship quality (Brown et al., 2022). This points to amplification rather than mismatch: when something negative is salient, whether an external event in daily life or a disagreement in the lab, shared NA appears to amplify rather than offset the cost of that context. This is closer to crossover accounts of stress transmission between partners (Westman, 2001) and to theoretical work on maladaptive coregulation in conflict (Butler & Randall, 2013).

Secondary findings

The exploratory response surface analyses (Appendix C) added two clarifications about the

similarity index itself: shared elevation, not symmetric proximity, drives the actual-affect surfaces, and the mismatch penalty built into the index finds little support in the observed data range. The preregistered effects are therefore best read as elevation effects (Edwards, 2002; Nestler et al., 2019). Further, trait we-ness in SC2 functioned as a relational resource in two complementary ways. As a main effect, higher we-ness predicted greater post-interaction love and closeness, after the positive but also after the negative interaction, suggesting that couples with stronger shared coping recover more positively from conflict regardless of how aligned their negativity was. As a moderator, trait we-ness produced a single significant interaction. At high we-ness, perceived shared NA was associated with greater closeness after a disagreement, while at low we-ness it predicted less closeness. Both patterns are consistent with the SC1 responsiveness story, in which felt understanding reframes the relational meaning of shared negativity (Reis & Gable, 2015), but, with only one significant interaction, we do not interpret it further. Finally, the preregistered sensitivity refits (random slopes, time-trend covariates, individual-affect controls) and the additional ceiling-restricted refits left fixed-effect conclusions intact; details are reported in Appendix A.

Limitations

Four limitations warrant emphasis. First, the love measure has a structural ceiling: roughly 4% of beeps in SC1 sit at or above 100, and the full beep distribution is left-skewed. Diagnostics in Appendix A identify the source of the resulting heteroscedasticity and tail deviations in this ceiling. Ceiling-restricted refits leave fixed-effect conclusions stable, but a less compressed scale would offer more interpretive room, particularly for high-baseline-love couples. More broadly, ceiling and floor constraints are well-documented issues for bounded self-report measures and can bias longitudinal inferences (Dejonckheere et al., 2022; Wang et al., 2009): a participant who already reports very high (or very low) affect has limited room to deviate further, which mechanically compresses the similarity indices derived from those deviations. One way to address this in future research would be to compute range-normalised deviations before constructing the dyadic elevation and mismatch indices, rescaling each person's deviation by their available range above (100 – baseline) or below (baseline), and then applying the same similarity formula to those normalised values (see, e.g., Cohen et al., 1999).

Second, the preregistered event-valence index inherits the branching structure of the ESM protocol: it is defined only at moments without partner contact. The exploratory recovery of the moderator across the full beep set is informative, but it is not a clean preregistered test of mismatch. The empirical pattern of amplification-by-negativity should therefore be treated as a hypothesis worth confirming in studies that do not condition the moderator on a relational sub-context.

Third, the SC2 sample size (98 dyads) limits the power available to the interaction-level mediation tests and to the actual $\times$ perceived interactions tested in Model 1c. The SC2 mediation null and the absence of SC2 actual $\times$ perceived interactions should not be read as evidence that perception plays no role at this scale. Rather, they are statements about the resolution at which interaction-averaged data can detect such patterns.

Fourth, the sample composition limits generalisation. Recruitment partly proceeded through psychiatric and psychotherapy clinics, and a substantial proportion of participants reported current medication use. This is a feature of the larger Re-Connect project rather than a flaw of the present study. However, it should temper extrapolation to community samples in which baseline distress, support-seeking, and dyadic-coping resources may differ. The sample was also predominantly cohabiting and predominantly White Belgian/Dutch, limiting cross-cultural inference.

Conclusion

PA similarity behaved much as predicted: shared positivity tracked momentary love, largely through perceived similarity, and responsiveness substituted for that signal when partners already felt understood. NA similarity told a more textured story: aversive in the moment, affiliative at the between-couple level, harmful most sharply when something negative was already happening, and attenuated when responsiveness or trait we-ness was high. The two valences are not mirror-image phenomena across all timescales and contexts, and the field will benefit from treating them separately and from being explicit about the level at which similarity is measured. A related observation across both components is that, in our data, perceived similarity behaved as a momentary mechanism: it carried most of the within-moment similarity–love association in SC1, while the analogous mediation aggregated across an 8-min interaction in SC2 was not detectable.

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Appendix A.

Sensitivities and Robustness Checks

This appendix documents the diagnostic checks and the sensitivity analyses for the Study Component 1 main-effects models (Models 1a and 1b, covering Hypotheses H1.1, H1.2 and H1.3). Because Models 1c, 2, 3a and 3b share the same bounded outcome and multilevel structure, the diagnostic concerns and the conclusions drawn from the sensitivities below extend to those models. Diagnostic analyses for the Study Component 2 outcomes are summarised at the end.

Diagnostic Checks

For each main-effects model, we inspected normalised residuals against fitted values, normal QQ-plots, and the alignment of fitted values with the 0–100 outcome bound.

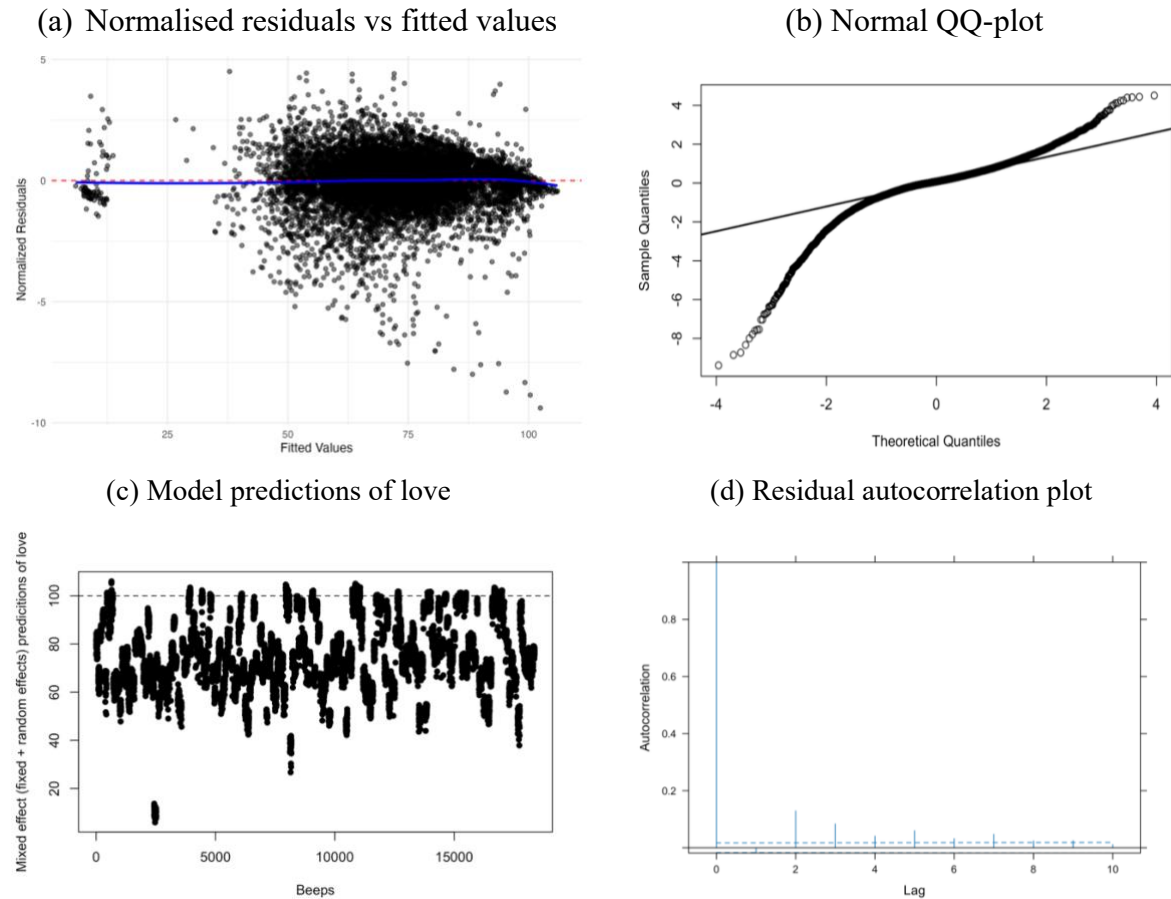
Three patterns emerged consistently. First, residual variance compresses towards higher fitted values, producing visible heteroscedasticity, while the LOESS smoother in the same panel stays close to zero across the fitted range and supports approximate linearity (Figure A1, panel (a)). Second, the QQ-plots show approximate normality between roughly ± 1.5 theoretical quantiles, with departures concentrated in the tails, especially the left tail (Figure A1, panel (b)). Third, approximately 4.4% of predictions for H1.1 sit at or above the scale ceiling of 100 (Figure A1, panel (c)). All three patterns reflect the structural ceiling on the love measure (Figure A2). The bounded scale necessarily truncates positive residuals when fitted values approach 100: a participant with a fitted value of 85, for example, can produce a positive residual of at most +15 but a negative residual as large as –85. This asymmetry is structural rather than diagnostic of misspecification.

Four considerations support fixed-effect inference under these conditions. First, the structural origin of the departures means they do not indicate that the model has missed a systematic source of variation. Second, the continuous AR(1) structure absorbs most short-term temporal dependence, with only minor residual autocorrelation remaining at short lags (Figure A1, panel (d)). Third, the preregistered ceiling-restricted sensitivity analyses (Table A2) excluded dyads in which at least one partner contributed fewer than 5, 10, or 20 non-ceiling responses and yielded consistent fixed-effect estimates across all three progressively stricter thresholds. Finally, given the sample size (13,302

beeps from 102 dyads), the central limit theorem provides additional support for fixed-effect inference under non-normality (Hox et al., 2017).

Figure A1

Model Diagnostic Plots for Study Component 1 (Model 1a: Actual PA Similarity)

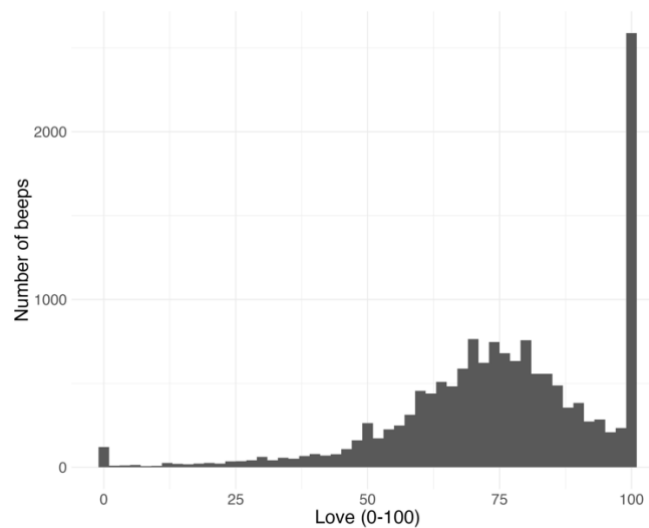


Note: Diagnostic plots shown for Model 1a (H1.1, actual PA similarity). Panels (a), (b), and (d) were inspected for all main-effects models (1a and 1b) and showed the same pattern. Panel (c) was computed for Model 1a; the structural ceiling argument extends to the other love-outcome models (1c, 2, 3a, 3b), which share the same bounded outcome and multilevel structure. Panel (a) shows residual variance compressing at high fitted values, reflecting the ceiling effect on the love scale. The LOESS smoother (blue) tracks the zero line closely across the fitted range, supporting approximate linearity. Panel (b) shows approximate normality within ± 1.5 theoretical quantiles, with heavier tails. Panel (c) shows mixed-effect predictions plotted across beeps; the dashed line marks the scale ceiling (100), and approximately 4.4% of predictions fall at or above this bound. Panel (d) shows residual

autocorrelation by lag; the continuous-time AR(1) structure absorbs most short-term dependence, with only minor residual autocorrelation at short lags.

Figure A2

Distribution of Momentary Love Ratings Illustrating the Ceiling Effect



Note. The histogram shows the distribution of momentary love ratings across all 13,302 ESM observations. A non-trivial proportion of ratings cluster at or near the maximum (100), consistent with the heteroscedasticity and fitted-value truncation visible in Figure A1. Ceiling-restricted sensitivity analyses (Table A2) excluded dyads in which at least one partner had fewer than 5, 10, or 20 non-ceiling responses; fixed-effect conclusions held across all three thresholds.

Preregistered Sensitivities

The preregistration specified three robustness checks: (1) refitting the primary models with random slopes for the within-level similarity predictors, (2) adding a linear time-trend covariate to evaluate drift across the ESM period, and (3) adding the actor's and partner's individual affect levels as covariates. Table A1 reports the model coefficients under the primary specification and under each preregistered sensitivity.

The random-slope refits leave the fixed-effect conclusions intact across all models. Signs and significance are preserved, and point estimates shift only modestly. Slopes vary meaningfully across dyads and persons, but their averages remain close to the primary fixed-slope estimates, supporting the interpretation of the primary coefficients as average effects rather than uniform slopes.

The time-trend covariate, tested for H1.1 on top of the random-slope structure, is statistically significant but practically negligible ($\beta = 0.009$ per hour, 95% CI [0.006, 0.012], $p < .001$), corresponding to roughly 3 points of upward drift over the full ESM period. The similarity coefficients were unchanged.

Table A1

Coefficients for the Main-Effects Models Under the Primary Specification and Preregistered Sensitivities for Study Component 1

Hypothesis / Coefficient	Primary	+ Random Slopes	+ Individual Affect Levels
H1.1: PA actual → love			
Within (β_1)	0.113***	0.134***	-0.053***
Between (β_2)	0.214	0.456	0.192
H1.2: NA actual → love			
Within (β_1)	-0.152***	-0.128***	-0.036***
Between (β_2)	0.621**	0.469*	0.629**
H1.3 (PA): actual + perceived → love			
Actual within (β_1)	0.032***	0.042***	-0.025**
Actual between (β_2)	-0.355	-0.191	-0.363
Perceived within (β_3)	0.149***	0.165***	-0.039***
Perceived between (β_4)	0.859***	0.913***	0.852***
H1.3 (NA): actual + perceived → love			
Actual within (β_1)	-0.094***	-0.081***	-0.038***
Actual between (β_2)	0.010	-0.130	0.016
Perceived within (β_3)	-0.127***	-0.119***	0.009
Perceived between (β_4)	0.893***	0.882***	0.896***

Note. Numbers are unstandardised fixed-effect estimates. "+ Random slopes" adds dyad- and person-level random slopes for the within-level similarity predictors. "+ Individual affect levels" adds actor and partner individual affect as covariates; these coefficients reflect the residual mismatch component of similarity once elevation has been absorbed and are not directly comparable in magnitude to the primary estimates. The time-trend sensitivity is not shown; results were consistent with the primary estimates. Bold = $p < .05$. * $p < .05$, ** $p < .01$, *** $p < .001$.

Because the similarity index combines the dyadic mean (elevation) of the two partners' affect with a mismatch penalty, including the individual affect levels absorbs the elevation component,

leaving only the residual mismatch component. The attenuated coefficients should therefore not be read as evidence against the similarity effect; they confirm that the residual mismatch component is itself associated with love. For NA, the residual within-dyad coefficient remains negative and significant in both H1.2 and H1.3. The perceived similarity between-coefficients are essentially unchanged, indicating that stable between-person differences in perceived similarity carry information that is not reducible to individual affect levels.

For PA, the residual within-dyad coefficient turns slightly negative in H1.1 ($\beta_I = -0.053, p < .001$) and H1.3 ($\beta_I = -0.025, p < .01$). Two factors account for this sign-flip. First, scale compression near the love ceiling modestly amplifies it: refitting Model 1a for H1.1, for example, on observations with love < 85 attenuates β_I to -0.030 and reduces its significance, but does not eliminate it. Second, and more substantively, the symmetric absolute-difference term in the similarity index masks a directional asymmetry. Replacing the absolute difference with the signed actor–partner PA difference yields $\beta = +0.075$ ($p < .001$): love increases when the actor's PA exceeds the partner's, and decreases when the partner's PA is higher. Averaging these opposite directional effects into a single symmetric penalty produces the negative residual coefficient. The response surface analyses in Appendix C provide direct support for this directional pattern. The negative β_I are therefore best understood as an artefact of the absolute-value formulation in the similarity index, modestly exacerbated by scale compression, not as evidence that PA similarity reduces love.

Additional Refits on Ceiling-Restricted Subsamples

Because the structural ceiling on love drives the structural patterns, we refitted the H1.1 model on three subsamples that progressively exclude dyads in which at least one partner contributed no more than 5, 10 or 20 non-ceiling beeps (Table A2). The within-dyad association is stable and highly significant across all subsets. The between-dyad coefficient strengthens as ceiling-dominated dyads are removed, becoming significant in the most restrictive subsample. Dyads whose love ratings are near the maximum carry little informative variability, and removing them sharpens rather than weakens the between-dyad estimate. Equivalent refits were not run for H1.2 and H1.3 because the structural source of the diagnostic concern is shared.

Table A2

H1.1 Coefficients (PA Actual Similarity → Love) on Ceiling-Restricted Subsamples

Subsample	Dyads	β_1 within	β_2 between
Primary (full sample)	102	0.113*** [0.104, 0.122]	0.214 [−0.314, 0.742]
Excl. dyads with ≤ 5 non-ceiling beeps	92	0.120*** [0.109, 0.130]	−0.040 [−0.546, 0.467]
Excl. dyads with ≤ 10 non-ceiling beeps	86	0.131*** [0.120, 0.142]	0.438 [−0.080, 0.956]
Excl. dyads with ≤ 20 non-ceiling beeps	77	0.140*** [0.129, 0.152]	0.820** [0.297, 1.343]

Note. A dyad was excluded if at least one partner had fewer than the indicated number of love ratings below 100, computed across the full ESM period. Bold = $p < .05$. ** $p < .01$, *** $p < .001$.

Study Component 2 Diagnostics

Diagnostics for the four H1.1 main-effects models in Study Component 2 (actual PA similarity predicting post-interaction love and closeness, separately within the positive and negative interaction contexts) show the same as in Study Component 1: heteroscedasticity from ceiling compression, a mild downward dip in the LOESS smoother at lower fitted values where observations are sparse, and left-skewed residuals in the QQ-plots. The closeness models are somewhat better-behaved than the love models, but in all four cases the central body of the residual distribution tracks the reference line adequately, and the structural source of the ceiling-driven departures is shared with Study Component 1. The same structural concerns apply to the H1.2 and H1.3 models, which share the bounded outcome.

Summary

The three preregistered Study Component 1 sensitivities (random slopes, time-trend covariate, and individual-affect adjustments) and the additional ceiling-restricted refits leave the fixed-effect conclusions intact in direction and magnitude. Study Component 2 shares the same ceiling-driven diagnostic pattern. The bounded love measure remains the dominant residual source of imperfect model fit, but the substantive inferences in the body are reliable.

Appendix B.

Exploratory moderation analyses with alternative operationalisations of event valence

The preregistered event-valence index C was presented only at beeps on which the participant was neither with their partner nor in recent contact, restricting H3.3/H3.4 tests to a narrow subset of moments. We re-estimated the same mixed-effects moderation models under three exploratory alternatives that recover substantially more beeps, replacing only the moderator. These alternatives additionally draw on positive and negative partner-specific event items presented in the complementary branches of the beep tree (asking whether something positive or negative had happened between the partners). Table B1 reports the four similarity $\times$ moderator interaction coefficients for each alternative.

Table B1

Interaction Term Coefficients for Exploratory Event-Valence Moderation Alternatives (H3.3/H3.4)

Alternative Moderators	PA Similarity		NA Similarity	
	Actual	Perceived	Actual	Perceived
C^{expl}	-0.049*** [-0.060, -0.038]	-0.037*** [-0.047, -0.026]	0.055*** [0.039, 0.071]	0.053*** [0.038, 0.068]
D^{pos}	-0.036*** [-0.055, -0.018]	-0.004 [-0.025, 0.017]	0.039* [0.009, 0.070]	0.010 [-0.021, 0.041]
D^{neg}	0.128*** [0.101, 0.155]	0.118*** [0.092, 0.144]	-0.127*** [-0.160, -0.093]	-0.147*** [-0.178, -0.115]

Note. Numbers are momentary similarity $\times$ moderator interaction coefficients with 95% CIs in

brackets. Moderators: C^{expl} = extended continuous valence index (person-mean centred; higher = more positive moments); D^{pos} = binary positive-event indicator (1=positive moment); D^{neg} = binary negative-event indicator (1=negative moment). Coefficients capture how the similarity–love slope changes per unit of C^{expl} , or between focal moments ($D = 1$) and other moments ($D = 0$). For PA, a positive coefficient = the association strengthens in the focal direction (more positive C^{expl} ; positive moments for D^{pos} ; negative moments for D^{neg}); a negative coefficient = the association weakens. For NA, interpretation reverses. * $p < .05$. *** $p < .001$.

Alternative 1: extended continuous valence index (C^{expl}). C^{expl} was constructed as a net-balance index in the same way as C , but uses the partner-specific positive and negative event items for

the beeps where the general event item set is missing. All four interactions were significant ($ps < .001$). For PA similarity, the interactions were negative: simple slopes remained positive throughout but attenuated as moments became more positive than typical, consistent with the preregistered mismatch direction (H3.3). For NA similarity, the interactions were positive and of comparable magnitude, meaning the negative association between NA similarity and love was strongest (most negative) at more-negative-than-typical moments, opposite to the mismatch prediction (H3.4).

Alternative 2: positive-event dummy (D^{pos}). D^{pos} was coded as 1 when the positive general or partner-specific item (whichever is available) is rated ≥ 2 , and 0 otherwise. Only the two actual similarity interactions were significant. The $PA \times D^{pos}$ interaction was negative: the PA similarity–love association was weaker on positive-event beeps, consistent with H3.3's prediction that PA similarity is most beneficial in less-positive moments. The $NA \times D^{pos}$ interaction was positive: the negative NA similarity–love association attenuated on positive-event beeps, contradicting H3.4: NA similarity was less harmful when something positive had occurred.

Alternative 3: negative-event dummy (D^{neg}). D^{neg} is coded as 1 when the negative general or partner-specific item (whichever is available) is rated ≥ 2 , and 0 otherwise. All four interactions were significant ($ps < .001$) and were substantially larger than under previous alternatives. The positive PA similarity–love association was roughly two to three times stronger on negative-event beeps than on other beeps (consistent with H3.3: PA similarity most beneficial in adverse moments). The negative NA similarity–love association was likewise two to three times stronger on negative-event beeps (contradicting H3.4: shared NA was more harmful, not less, when something negative had occurred).

Convergence across the three alternatives. The preregistered null is largely explained by the restricted sample on which C was defined: when the moderator is constructed on the full beep set, all four interactions emerge consistently across alternatives. For PA similarity, the association with love tended to be stronger in less-positive moments, consistent with H3.3. For NA similarity, all three alternatives contradicted H3.4: shared NA was most detrimental in negative, not positive, moments, a pattern more consistent with amplification than with the preregistered mismatch account.

Appendix C.

Response Surface Analysis of Affect Similarity and Momentary Love (Exploratory)

The similarity index in Study Component 1 imposes implicit assumptions about the shape of the actor $\times$ partner affect surface, including the assumption that both directions of mismatch are equally detrimental. Response surface analysis (RSA; Edwards, 2002; Nestler et al., 2019) offers a complementary perspective by fitting a second-degree polynomial regression that leaves the functional form of the association free, allowing the data to reveal the shape of the joint actor–partner affect surface.

Method

Four models were estimated, one for each combination of affect valence (PA, NA) and partners' affect source (actual versus perceived). Each model took the form:

$$L_{mpd} = b_0 + b_1X_{mpd} + b_2Y_{mpd} + b_3X_{mpd}^2 + b_4X_{mpd}Y_{mpd} + b_5Y_{mpd}^2 + \varepsilon_{mpd}$$

where L is momentary love, X is the actor's person-mean centred affect, and Y is either the partner's person-mean centred self-reported affect (actual models) or the actor's person-mean centred perception of the partner's affect (perceived models). The subscripts m , p , and d denote moments, persons, and dyads, respectively. All models included random intercepts for dyads and persons within dyads, and a CAR(1) residual correlation structure, consistent with the Study Component 1 specifications.

Following Edwards (2002) and Nestler et al. (2019), the polynomial coefficients b_1 – b_5 were transformed into five response surface parameters:

1. **a_1 (LOC slope) = $b_1 + b_2$:** slope along the line of congruence ($X = Y$), reflecting shared-elevation effects. A rising ridge pattern ($a_1 > 0$ for PA, $a_1 < 0$ for NA) means love tracks the shared affect level.
2. **a_2 (LOC curvature) = $b_3 + b_4 + b_5$:** curvilinearity along the LOC. Negative values indicate that shared affect has diminishing (or reversing) returns.
3. **a_3 (LOIC slope) = $b_1 - b_2$:** slope along the line of incongruence ($X = -Y$), reflecting directional asymmetry. A non-zero a_3 means that actor-exceeds-partner and partner-exceeds-actor

mismatches have different associations with love.

4. a_4 (**LOIC curvature**) = $b_3 - b_4 + b_5$: curvature along the LOIC. A negative a_4 (inverted-U) would indicate a symmetric congruence penalty: love decreases as actor and partner affect diverge, regardless of direction. This is the defining signature of strict congruence effects.
5. a_5 (**FPA alignment**) = $b_3 - b_5$: displacement of the first principal axis (ridge) from the LOC. If $a_5 \neq 0$, the optimal partner affect for maximising love is not identical to the actor's own affect.

Results

Table C1 summarises the response surface parameters for all four models.

Actual PA Surface. The actual PA surface is characterised by a strong positive LOC slope ($a_1 = 0.310, p < .001$), indicating a rising ridge: predicted love increases as both partners' PA rise together. This elevation component closely matches the elevation coefficient from the exploratory signed-difference decomposition reported in Appendix A ($\beta = 0.29$). A positive LOIC slope ($a_3 = 0.178, p < .001$) indicates directional asymmetry: moments when the actor's PA exceeds the partner's predict higher love than moments of the reverse pattern. This directly corroborates the directional finding from Appendix A, where replacing $|PA_{own} - PA_{part}|$ with the signed difference ($PA_{own} - PA_{part}$) yielded $\beta = +0.075 (p < .001)$. The RSA makes the same pattern visible from the surface geometry without requiring a separate decomposition, because the polynomial imposes no symmetric constraint on actor and partner PA. Crucially, a_4 is positive (0.005), not negative, providing no evidence for a symmetric congruence penalty. Both curvature parameters are an order of magnitude smaller than the LOC and LOIC slopes and have a negligible practical impact within the observed data range.

Actual NA Surface. The actual NA surface is structurally analogous to the PA surface with the signs reversed. A negative LOC slope ($a_1 = -0.287, p < .001$) confirms a descending ridge: shared higher NA predicts lower love moment to moment. A negative LOIC slope ($a_3 = -0.131, p < .001$) indicates that moments when the actor's NA exceeds the partner's are particularly associated with lower love, the directional counterpart of the PA pattern. As with PA, a_4 is positive (0.003), not negative. The NA surface within the observed range is well described as a downward-tilted plane, not a symmetric congruence surface.

Table C1

RSA surface parameters across all four models.

Model		Response surface parameters				
Actor	Partner	<i>a</i> ₁ (LOC slope)	<i>a</i> ₂ (LOC curv.)	<i>a</i> ₃ (LOIC slope)	<i>a</i> ₄ (LOIC curv.)	<i>a</i> ₅ (FPA = LOC)
Actual PA	Actual PA	0.310***	-0.001***	0.178***	0.005***	0.001***
	Perceived PA	0.385***	-0.000	-0.012	0.004***	0.000
Actual NA	Actual NA	-0.287***	-0.002***	-0.131***	0.003***	0.000
	Perceived NA	-0.310***	-0.000	-0.072***	0.001	-0.000

Note. LOC = line of congruence ($X = Y$); LOIC = line of incongruence ($X = -Y$); FPA = first

principal axis. Parameters are derived from second-degree polynomial multilevel models with random intercepts for dyads and persons, and continuous-time AR(1) residuals. All predictors are person-mean centred. Bold = $p < .05$. *** $p < .001$.

Perceived PA Surface. The perceived PA surface shows a steeper LOC slope than the actual PA surface ($a_1 = 0.385$ vs 0.310 , both $p < .001$), consistent with the Study Component 1 H1.3 finding that perceived PA similarity carries a stronger within-level association with love than actual PA similarity. Notably, the directional asymmetry disappears: $a_3 = -0.012$, ns. When the partner's affect enters the model as perceived by the actor rather than as an objective self-report, the surface becomes approximately symmetric around the LOC.

Perceived NA Surface. The perceived NA surface shows the steepest descending ridge across all four models ($a_1 = -0.310$, $p < .001$), again consistent with H1.3's finding that perceived NA similarity is more strongly negatively associated with love than actual NA similarity. A small but significant negative LOIC slope persists ($a_3 = -0.072$, $p < .001$). Neither curvature parameter reaches significance for perceived NA. The surface is well characterised as a steeply descending tilted plane.

Cross-Model Interpretation

Three patterns emerge consistently across the four models. First, elevation dominates. The LOC slopes are the largest parameters in every model, and perceived-affect LOC slopes exceed their actual-affect counterparts (actual PA: 0.310 vs. perceived PA: 0.385 ; actual NA: -0.287 vs. perceived NA: -0.310). This reproduces the H1.3 finding within the RSA framework and confirms that shared

affect elevation, whether objective or perceived, is the primary driver of the actor–partner affect–love association.

Second, directional asymmetry is a feature of actual affect, not perceived affect. The LOIC slopes are substantial for actual PA ($a_3 = 0.178$) and actual NA ($a_3 = -0.131$) but attenuated or absent for the perceived models. Substantively, when partner affect enters the model objectively, the surface tilts toward the actor: having more positive (or less negative) affect than the partner is associated with higher love. This actor-favouring directional effect is fully absorbed when partner affect is replaced by the actor's perception of it, as perceptions are already more aligned with the actor's own state.

Third, no model produces a symmetric congruence penalty. An inverted-U on the LOIC ($a_4 < 0$) would indicate that love decreases as partners' affects diverge, irrespective of direction. No such pattern appears. Where a_4 reaches significance it is positive, and in all cases it is an order of magnitude smaller than a_1 . This finding has a direct implication for interpreting the preregistered similarity index: the composite S index builds in a symmetric mismatch penalty via the $|X - Y|$ term, but the RSA surfaces do not support this component. The preregistered effects are therefore best attributed to shared elevation rather than to congruence per se.

Limitations

These analyses were not preregistered and should be interpreted as exploratory. Additionally, because all predictors are person-mean centred, the RSA surfaces reflect within-person (momentary) associations only and do not capture between-dyad effects. A full multilevel RSA separating within- and between-level surface parameters (Nestler et al., 2019) would require including dyad-mean terms as Level-2 predictors and is a possible extension for future work. Finally, the models include only random intercepts; adding random slopes for the five polynomial terms would be computationally demanding and likely under-identified in the present sample.

Appendix D.

Operationalisation of Trait We-ness

To operationalise trait-level we-ness, we used the available data of the Dyadic Coping Inventory (DCI) (Bodenmann, 2008; Ponnet, 2012). Because only 30 DCI items were available in the pre-existing data, we could not include common dyadic coping (items 31–35) or evaluation items (36–37). For each person, a trait we-ness score was calculated as the mean of the 22 items reflecting constructive stress communication and supportive dyadic coping by self and partner (supportive and delegated):

1. Stress Communication (Self & Partner):

Items 1, 2, 3, 4, 16, 17, 18, and 19, reflecting the open expression of stress and feelings.

2. Supportive Dyadic Coping (Perceived Partner Support):

Items 5, 6, 8, 9, 12, 13, and 14, reflecting emotion-focused and problem-focused support received from the partner.

3. Supportive Dyadic Coping (Own Provided Support):

Items 20, 21, 23, 24, 28, 29, and 30, reflecting emotion-focused and problem-focused support provided to the partner.

We thus excluded items 7, 10, 11, 15, 22, 25, 26, and 27, related to negative dyadic coping (hostile, ambivalent, or superficial coping), to ensure the score represents positive ‘we-ness’.